

OMER GADALLA

Aerospace Engineer

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EDUCATION

Middle East Technical University (METU) — (2021–2026)

B.Sc. Aerospace Engineering

EXPERIENCE

Galaxy Rocket Team — (Dec 2025-Present)

Aerodynamics Engineer

TEKNOFEST Liquid Bipropellant Rocket Competition (A7) | 2026

- Contribute to the design of a **liquid bipropellant rocket** targeting total impulse of 1200 N.s, apogee of 3000 ft and PMF of 0.3 based on the competition requirements.
- Perform external **aerodynamic CFD** to compute drag coefficients and aerodynamic loads.
- Execute **injector and combustion chamber CFD** analyses.

Propulsion Engineer

TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) | 2025-2026

● Design & Analysis Phase

- Contributed to the design, analysis and tests of a **100N liquid bipropellant rocket engine**
- Modeled the **propellant feed system** using MATLAB Simscape.
- Conducted **combustion and CFD analyses** to investigate chamber flow.
- Performed **thermo-structural FEA** to assess structural integrity under **high-pressure** and **high temperature**.

● Testing Phase

- Assisted with **spray injector cold flow tests**, supporting evaluation of **spray pattern** and **atomization** behavior, and comparing test results with **CFD** simulation data.
- Collaborated with the team to do component-level **valve tests**, including **functional, performance, and safety** tests.
- Achieved **Finalist** ranking among national teams.

MTS Defense Industry — (July 2025-Sep 2025)

R&D Engineer – Intern

- Developed the **conceptual design** for a **20N green bipropellant thruster** for small satellite applications.
- Designed the **propellant feed system** using MATLAB Simscape.
- Proposed **the project** to the investors, highlighting system's technology and development risks.

Technical SKILLS

Programming

- MATLAB
- Simulink
- Simscape
- NASA CEA

CAD

- Autodesk Inventor
- FUSION 360
- SolidWorks

CFD

- Ansys Fluent
- OpenFOAM

Languages

- Arabic (Native)
- Turkish (Fluent)
- English (Fluent)

Structural Analysis

- Ansys Mechanical /Nastran